

Role of Sleep and White Matter in the Link Between Screen Time and Depression in Childhood and Early Adolescence

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IMPORTANCE With the widespread adoption of screen-based devices among adolescents, there is growing concern that more screen time could contribute to mental health problems such as depression. It is thus critical to identify potential mediating factors that could help explain this potential risk relationship. Recent evidence indicates that more screen time could impact sleep duration and brain structural connectivity (ie, white matter organization), which are critical for emotional health. Notably, sleep duration is a modifiable behavior that health care providers can easily target.

OBJECTIVE To identify the association between screen time during late childhood and depressive symptoms in early adolescence, and to investigate whether these associations are mediated by sleep duration and white matter organization.

DESIGN, SETTING, AND PARTICIPANTS This prospective study was conducted from January 2024 to June 2024. Data from the Adolescent Behavior Cognitive Development (ABCD) Study were used to identify clinical and neuroimaging characteristics of participants at late childhood (T1; defined as aged 9-10 years) and early adolescence (T2; defined as age 11-13 years). Children and their parent/caregiver were recruited across 21 US cities. Participants with no past/current psychiatric disorders at T1 were selected for analyses. Initial analyses were conducted in 2024 and finalized in February 2025.

MAIN OUTCOMES AND MEASURE Outcomes included screen time assessed using a self-report questionnaire, sleep duration assessed using the Munich Chronotype Questionnaire, and depressive symptoms characterized using the Child Behavior Checklist. Neurite orientation dispersion and density imaging and a tract profile approach were used to characterize the orientation dispersion index of 3 white matter tracts that are known to be implicated with depression: cingulum bundle, forceps minor, and uncinate fasciculus.

RESULTS Analyses included 976 participants (460 children [47.1%] were female, mean [SD] age was 9.9 [0.6] years at T1 and 11.9 [0.6] years at T2). Each additional hour of daily screen time at T1 was associated with a 0.12-point (95% CI, 0.04-0.20; $P = .008$) increase in Child Behavior Checklist depressive score at T2. Shorter sleep duration and worse cingulum bundle organization (greater orientation dispersion index) at T2 mediated 36.4% (95% CI, 18.2%-63.6%) of the association between more screen time and more depressive symptoms.

CONCLUSIONS AND RELEVANCE Results of this study show that more screen time in late childhood was associated with more depressive symptoms, potentially due to shorter sleep and worse white matter organization during early adolescence. These findings emphasize the importance of promoting healthy habits and balancing screen time with adequate sleep.

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Screen-based devices are now a ubiquitous part of adolescents' lives.¹ They enable connection with peers, provide entertainment, and facilitate discussions on sensitive topics.^{2,3} Time spent on screens is not necessarily harmful,² with some studies finding no or a weak negative relationship between screen time and youth mental health.⁴ However, the widespread adoption of screen-based devices has also raised concerns about their impact on adolescents' mental health.^{5,6} Although research continues to develop, studies investigating time spent on activities such as social media, watching videos, and playing video games suggest a potential risk relationship with depression, particularly during windows of developmental sensitivity such as early adolescence.⁷⁻¹⁰ However, the mechanisms underlying this phenomenon remain unclear. Recent studies suggest that screen time is associated with worse health outcomes, such as insufficient sleep and more sleep problems.^{11,12} Given the modifiable nature of sleep and its role in emotional well-being and brain development,¹³⁻¹⁵ understanding the interplay between these factors is essential for informing strategies to mitigate risks and improve adolescent well-being.

During adolescence, major sleep-circadian changes contribute to a delayed sleep phase.¹⁶ This makes adolescents vulnerable to external influences, such as screen time, which can result in insufficient sleep and poor sleep quality.^{15,17} Additionally, sleep is closely linked to mental health,¹⁸ with insufficient sleep and poor sleep quality increasing the risk of depressive symptoms during adolescence.¹⁹⁻²¹ However, a gap remains on how sleep, screen time, and depression are linked. The link between screen time and depression may involve indirect effects of sleep on the adolescent brain. Sleep is critical for healthy neurodevelopment,^{13,14} influencing the maturation of brain regions implicated with emotional regulation,^{22,23} with noteworthy associations during early adolescence involving sleep duration and brain gray and white matter.²⁴⁻²⁶ Considering that alterations in brain maturation are associated with an increased risk for mental health problems,²⁷ when screen time leads to shorter sleep, it may negatively impact brain regions responsible for emotional regulation, thereby increasing the risk of depression. However, research is still in the early stages due to the complex and dynamic changes in the adolescent brain. In this regard, brain white matter can offer a robust approach to investigate these relationships.

During brain development, white matter undergoes critical changes, including myelination and reorganization of neurites (axons and dendrites).²⁸ These changes start early and may persist into adulthood.^{29,30} Although these changes are susceptible to external influences (eg, concussion, physical activity),^{31,32} research on the effects of sleep and screen time is still limited.³³⁻³⁵ For instance, less regular sleep, poor sleep quality, more sleep problems, and greater screen time during childhood have been associated with lower fractional anisotropy (FA; an indirect index of fiber collinearity).³³⁻³⁶ However, these are findings from independent studies, and non-specificity of FA³⁷ makes it challenging to parse out specific components affected by sleep and screen time. Recent work using neurite orientation dispersion and density imaging (NODDI)²⁸ has shown that short sleep in early adolescence is associated with worse fiber coherence (orientation dispersion

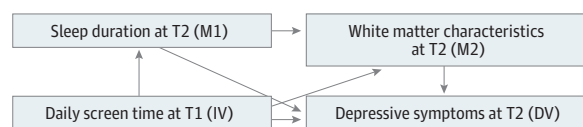
Key Points

Question Could the positive association between screen time and depression be explained by effects of screen time on sleep and white matter integrity among adolescents?

Findings In this study of 976 children, shorter sleep and worse white matter organization mediated 36.4% of the associations between more screen time and depressive symptoms in early adolescence. Additionally, shorter sleep mediated 37.5% of the association between more screen time and worse white matter integrity.

Meaning Sleep is a modifiable behavior and a key component of the effect of screen time on the brain and depression, underscoring the importance of strategies that encourage a balanced lifestyle to mitigate the negative impact of screen time on sleep.

Figure 1. Serial Mediation Framework



Proposed relationships involving screen time at T1 (late childhood), sleep duration at T2 (early adolescence), white matter characteristics at T2, and depressive symptoms at T2 are shown. Arrows indicate the direction of the proposed relationship. DV indicates dependent variable; M1, mediator 1; M2, mediator 2; IV, independent variable.

index [ODI])²⁶ of white matter tracts implicated with emotional regulation, such as the cingulum bundle,³⁸ forceps minor,³⁹ and uncinate fasciculus.⁴⁰ Given the sensitivity of NODDI to structural changes during development,⁴¹ these findings suggest a possible pathway in which screen time indirectly affects the adolescent brain and mental health through sleep. However, it remains unclear to what extent screen time could impact sleep and, in turn, white matter organization in regions implicated in emotion regulation, which could increase the risk for depression. A serial mediation model could help clarify direct and indirect pathways, providing insight into the underlying mechanisms linking screen time and depressive symptoms.

The present study aimed to identify the association between screen time during late childhood and depressive symptoms in early adolescence and examine whether these associations were mediated by sleep duration and white matter organization of emotion regulation tracts implicated with depression (cingulum bundle, forceps minor, and uncinate fasciculus). To this end, we leveraged the large longitudinal Adolescent Brain Cognitive Development (ABCD) Study using NODDI and tractography approaches to characterize these tracts at 2 points: late childhood (T1; defined as age 9-10 years) (mean [SD] age, 9.9 [0.6] years) and early adolescence (T2; defined as age 11-13 years) (mean [SD] age, 11.9 [0.6] years). Based on the existing literature described above, we proposed the framework illustrated in Figure 1. We hypothesized that (1) more daily screen time at T1 may predispose to more depressive symptoms at T2 and (2) that this relationship will be

Table 1. Demographic Characteristics of Participants at Study Entry (N = 976)

Demographic characteristics	No. (%)
Age, mean (SD), y	9.9 (0.6)
Sex	
Male	516 (52.9)
Female	460 (47.1)
Pubertal Development Scale Score, mean (SD)	1.7 (0.7)
Household income	
<\$5000	18 (1.9)
\$5000-\$11 999	21 (2.1)
\$12 000-\$15 999	7 (0.7)
\$16 000-\$24 999	38 (3.9)
\$25 000-\$34 999	49 (5.0)
\$35 000-\$49 999	82 (8.4)
\$50 000-\$74 999	135 (13.8)
\$75 000-\$99 999	152 (15.6)
\$100 000-\$199 999	346 (35.5)
>\$200 000	128 (13.1)
Parent education	
9th Grade	2 (0.2)
10th Grade	4 (0.4)
11th Grade	1 (0.1)
12th Grade	9 (0.9)
Associate degree: academic program	54 (5.5)
Associate degree: occupational	66 (6.8)
Bachelor's degree	300 (30.7)
Doctoral degree	56 (5.7)
GED or equivalent	14 (1.4)
High school graduate	41 (4.2)
Master's degree	272 (27.9)
Professional school degree	57 (5.9)
Some college	100 (10.3)
Race and ethnicity ^a	
Asian	14 (1.4)
Black	54 (5.5)
Hispanic	181 (18.6)
White	622 (63.7)
Other	105 (10.8)
Interview period ^b	
Summer	185 (19.0)
School year	791 (81.0)
Daily screen time, mean (SD), h ^c	3.2 (2.5)
Passive entertainment time	1.9 (1.5)
Active entertainment time	0.9 (0.9)
Communication time	0.4 (0.8)

^a In the ABCD study, parents reported race and ethnicity combined using the following options: Asian, Black, Hispanic, White, and other. Additional follow-up questions were done for race and ethnicity separately. For specific information about race and ethnicity as separate variables, see the eMethods and eTable 3 in [Supplement 1](#).

^b Summer includes July through August; school year includes September through June.

^c Calculated using time spent with 6 categories: watching TV, watching videos, texting, social media, video chatting, and playing video games. Categories of daily screen time represented time spent on passive entertainment (watching TV and watching videos), active entertainment (video games), and communication (texting, social media, video chat).

mediated by shorter sleep and worse fiber coherence (greater ODI) at T2. We explored how well tensor-based metrics (FA) captured the proposed associations in parallel to NODDI and examined the role of other sleep metrics and specific screen activities.

Methods

Participants

Data from the ABCD Study available on the National Institute of Mental Health Data Archive were used for the analyses. This is a multisite, longitudinal study designed to recruit more than 10 000 children ages 9 to 10 years and follow up over 10 years into early adulthood. All research protocols were approved by the institutional review board at each site. Parents or guardians and adolescents provided written informed consent and assent, respectively, to participate in the study.

The National Institute of Mental Health Data Archive includes tensor-based metrics (eg, FA), but it does not include NODDI metrics. Therefore, given the focus of this work on using NODDI, a computationally demanding process (eMethods in [Supplement 1](#)), raw diffusion magnetic resonance imaging data from the ABCD 5.1 release were used to derive the metrics of interest. A total of 1109 participants (47.5% female) with no history of psychiatric disorders at late childhood had all clinical and neuroimaging data required for analyses. Of the total sample, 133 participants were excluded due to poor quality of the postprocessing or outlier sleep/screen time data (>2 SDs from the mean), leaving 976 participants (47.1% females; [Table 1](#); [Figure 1](#) in [Supplement 1](#)) for statistical analyses across 2 time points. See eTable 1 in [Supplement 1](#) for scores and variables collected across follow-up visits. There were no between-group differences before and after accounting for quality control (eTable 2 in [Supplement 1](#)). This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline.

Assessments

Screen Time

Self-reported screen time was collected using a screen time questionnaire at both time points (T1 and T2).⁴² Daily use of 6 recreational categories (watching TV, watching videos, texting, using social media, videochatting, and playing videogames) was collected for weekdays and weekend. As done in previous studies,^{43,44} daily screen time was calculated based on a weighted mean of all categories ($[\text{weekday mean hours} \times 5] + [\text{weekend mean hours} \times 2]/7$). The following subcategories of daily screen time were derived for exploratory analyses: passive entertainment (watching TV, watching videos), active entertainment (playing videogames), and communication (texting, using social media, videochatting).

Sleep Duration

The sleep times collected with the Munich Chronotype Questionnaire⁴⁵ (only available at T2) were used to calculate self-reported sleep duration (the difference between sleep end and sleep onset in hours; sleep onset refers to the sum between

time preparing to sleep and sleep latency). Quality assurance identified and excluded outliers, inconsistencies, and illogical responses. Daily sleep duration was calculated based on a weighted mean of sleep duration on weekdays and weekends. Interview dates were used to categorize the timeframe (summer vs school year).

Depressive Symptoms

The Child Behavior Checklist⁴⁶ was used to characterize behavioral problems at T1 and T2. The Child Behavior Checklist is a parent-reported Likert-type questionnaire with 113 questions. Each question is scored using a 3-point scale (0 indicates absent; 1, occurs sometimes; and 2, occurs often; total score range, 0-238; higher scores indicate more symptoms) and can be grouped into the following *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition*-oriented scales: depressive, attention-deficit/hyperactivity disorder, anxiety, conduct, oppositional defiance disorder, and somatic behaviors.

Additional Information

Age, sex, pubertal status,⁴⁷ parent highest education level, household income, race, and ethnicity were collected.⁴⁸ Additional sleep metrics were calculated based on the Munich Chronotype Questionnaire: sleep timing, social jetlag, and weekly sleep loss. The parent-reported Sleep Disturbance Scale for Children⁴⁹ was used to determine total sleep problems and sleep duration at T1. See the eMethods and eTable 3 in [Supplement 1](#) for details.

Neuroimaging Data

Diffusion Metrics

Protocol and preprocessing steps are described in the eMethods in [Supplement 1](#). ODI maps were derived for each participant. Additionally, tensor-based (FA) and neurite density index maps were derived for each participant to identify their performance in capturing the hypothesized relationships. All processing steps were performed in native space. NODDI/FA maps were registered to the Montreal Neurological Institute space to extract metrics for analyses. Time between visits had no significant effect on T2 metrics (eMethods in [Supplement 1](#)).

White Matter Tract Segmentation

TractSeg^{50,51} was used to reconstruct the cingulum bundle,³⁸ forceps minor,³⁹ and uncinate fasciculus.⁴⁰ For each segmented tract, overall mean and nodal diffusion metrics were extracted for tract profile analyses (eMethods in [Supplement 1](#)).⁵² Given that laterality was not part of our hypotheses, we took the mean of the diffusion mean and nodal metrics across hemispheres for the cingulum bundle and uncinate fasciculus.

Harmonization

To account for the data being collected across different sites, ComBat, a batch-effect correction tool, was used to minimize this variance.⁵³ Age, sex, and Pubertal Development Scale score were included as covariates. There was no significant effect of site in any metric after harmonization (eTable 4 in [Supplement 1](#)).

Statistical Analysis

To test our main hypotheses, we used multivariate regressions. We used PROCESS macro⁵⁴ in R (R Foundation) to examine mediation effects and calculate direct, indirect, and total effects, along with regression coefficients, standard errors, and bootstrapped (10 000) CIs. Covariates in all models included sex at birth, pubertal timing, household income, parent education, and race and ethnicity. These covariates were selected for their social and biological relevance to adolescent development.⁵⁵⁻⁵⁸ Normality of data was confirmed using the Kolmogorov-Smirnov test. To account for multiple testing, the false discovery rate was used. Statistical testing was 2-sided, with $P \leq .05$ indicating statistical significance.

Screen Time and Depressive Symptoms

A model evaluated the effect of daily screen time at T1 (independent variable) on depressive symptoms at T2 (dependent variable). Depressive symptoms at T1 were included as an additional covariate.

Screen Time and Sleep Duration

A model evaluated the effect of daily screen time at T1 (independent) on sleep duration at T2 (dependent).

Sleep Duration and White Matter

Models evaluated the effect of sleep duration (independent) on diffusion metrics at T2 for each white matter tract of interest (cingulum bundle, forceps minor, and uncinate fasciculus; dependent variables). For those tracts in which sleep duration was significantly associated with mean diffusion metrics, tract profile analyses evaluated whether the association was focal (eg, frontal nodes) or widespread (all nodes). The interview timeframe was included as an additional covariate.

White Matter and Depressive Symptoms

Models evaluated the effect of diffusion metrics identified in the sleep duration and white matter models described above (independent) on depressive symptoms at T2 (dependent).

Serial Mediation Model

The mediated association between daily screen time at T1 (independent) and depressive symptoms at T2 (dependent) was tested using a serial mediation model. In this model, sleep duration was defined as mediator 1 (M1) and diffusion metrics identified in the sleep duration and white matter models described above were defined as M2. Additionally, associations between independent variables and M2 and between M1 and dependent variables were part of the model and helped to account for alternative relationships and mediation paths. We also evaluated whether the effect of the mediator on the dependent variable was consistent across levels of the exposure⁵⁹ and the percentage of the total effect explained by indirect effects.

Exploratory Analyses

Analyses included sensitivity models; inverse relationships involving screen time, sleep duration, and depressive symptoms; the relationships involving other behavioral problems,

Table 2. Effect of Sleep Duration on Diffusion Metrics

White matter tracts (dependent variables)	β (95% CI)	P value	Q ^a
Neurite density index (z score)			
Cingulum bundle	−0.003 (−0.07 to 0.06)	.90	.90
Forceps minor	−0.02 (−0.08 to 0.05)	.59	.90
Uncinate fasciculus	−0.02 (−0.09 to 0.05)	.60	.90
Orientation dispersion index (z score)			
Cingulum bundle	−0.11 (−0.17 to −0.04)	.001	.005
Forceps minor	−0.01 (−0.08 to 0.05)	.66	.66
Uncinate fasciculus	−0.06 (−0.12 to 0.01)	.07	.11
Fractional anisotropy (z score)			
Cingulum bundle	0.08 (0.01 to 0.14)	.02	.06
Forceps minor	0.04 (−0.02 to 0.11)	.20	.29
Uncinate fasciculus	0.001 (−0.06 to 0.07)	.97	.97

^a P values after false discovery rate correction for multiple comparisons.

sleep metrics, and screen time modalities; screen time thresholds for significant effects; and the influence of family membership. See the eMethods in [Supplement 1](#) for details of exploratory analyses.

Results

A total of 976 participants (460 [47.1%] children were female; mean [SD] age, 9.9 [0.6] years at T1 and 11.9 [0.6] years at T2) were included in the analyses.

Screen Time and Depressive Symptoms

Each 1-hour increase in daily screen time at T1 was associated with a 0.12-point (95% CI, 0.04-0.20; $P = .008$) increase in Child Behavior Checklist depressive symptoms at T2.

Screen Time and Sleep Duration

Each 1-hour increase in T1 daily screen time was associated with a 0.21-point decrease in T2 sleep duration (95% CI, −0.28 to −0.14; $P < .001$).

Sleep Duration and White Matter

Participants reported a mean (SD) of 9.0 (1.0) hours of daily sleep (range, 6.1-12.0 hours). Each 1-hour decrease in sleep duration was associated with a 0.11-point increase in cingulum bundle ODI (95% CI, −0.17 to −0.04; $P = .001$; [Table 2](#)). More specifically, short sleep was associated with low middle cingulum bundle ODI ([Figure 2](#); [eTable 5](#) in [Supplement 1](#)).

White Matter and Depressive Symptoms

Each unit increase in the middle cingulum bundle ODI was associated with a 0.15-point increase in depressive symptoms at T2 (95% CI, 0.04-0.25; $P = .01$).

Serial Mediation Model

Shorter sleep at T2 mediated 37.5% (95% CI, 12.5%-50.0%) of the effects of longer daily screen time at T1 on worse fiber coherence in the middle cingulum bundle at T2. Worse fiber

coherence in the middle cingulum bundle at T2 mediated 12.5% (95% CI, 2.0%-22.7%) of the effect of shorter sleep at T2 on more depressive symptoms at T2. In the complete model, shorter sleep and worse fiber coherence in the middle cingulum bundle mediated 36.4% (95% CI, 18.2%-63.6%) of the effects of longer daily screen time on more depressive symptoms ([Figure 3](#); [eResults](#) in [Supplement 1](#)).

Exploratory Analyses

Screen time thresholds for significance included 1 hour for sleep duration (0.34-point decrease [95% CI, −0.53 to −0.16]), 2 hours for depressive scores (0.14-point increase; [95% CI, 0.03-0.27]), and 3 hours for cingulum bundle ODI (0.16-point increase [95% CI, 0.04-0.29]). Additionally, longer parent-reported sleep duration at T1 was associated with more screen time at T2 ($F = 19.05$; $P < .001$). See [eTable 6-7](#) and [eFigure 2-3](#) in [Supplement 1](#).

Discussion

Findings from this study support the first hypothesis, indicating that more daily screen time in late childhood was associated with more depressive symptoms in early adolescence. Additionally, supporting the second hypothesis, short sleep and worse cingulum bundle coherence mediated these effects. Exploratory analyses revealed a dose-dependent pattern in which 1 hour of daily screen time was associated with shorter sleep, 2 hours with more depressive symptoms, and 3 hours with worse cingulum bundle organization.

Several mechanisms help explain the effect of screen time on sleep. Blue light from screens may disrupt melatonin, circadian rhythms, and sleep timing.^{60,61} Additionally, evening screen use can displace sleep⁶¹ and stimulating content can delay bedtime.^{62,63} Based on that, limits on screen use have been proposed to prevent adverse outcomes.⁶⁴ However, strict restrictions are impractical, especially for adolescents who rely on screens for other activities such as homework.⁶⁵ Moreover, the efficacy of these restrictions remains unclear, because evidence suggests that screen light has small-to-negligible effects on sleep.⁶⁶ Although the absence of the Munich Chronotype Questionnaire at T1 limited our ability to examine temporal dynamics, exploratory analyses also indicated a potential bidirectional relationship between screen time and sleep duration in early adolescence, requiring further study of causal relationships. Altogether, the findings show that screens pose potential risks, but engaging parents, educators, and adolescents is essential to develop balanced strategies addressing screen use and sleep based on individual needs.

Although previous studies have shown associations involving screen time, sleep, and depression in older adolescents and young adults,^{11,67,68} to the best of the authors' knowledge, the current findings are the first to show that white matter is important in these relationships during early adolescence. This underscores the importance of considering specific neural mechanisms when studying the impact of screen time on mental health in younger adolescents. These findings also suggest distinct pathways depending on the screen time

Figure 2. Cingulum Bundle Findings

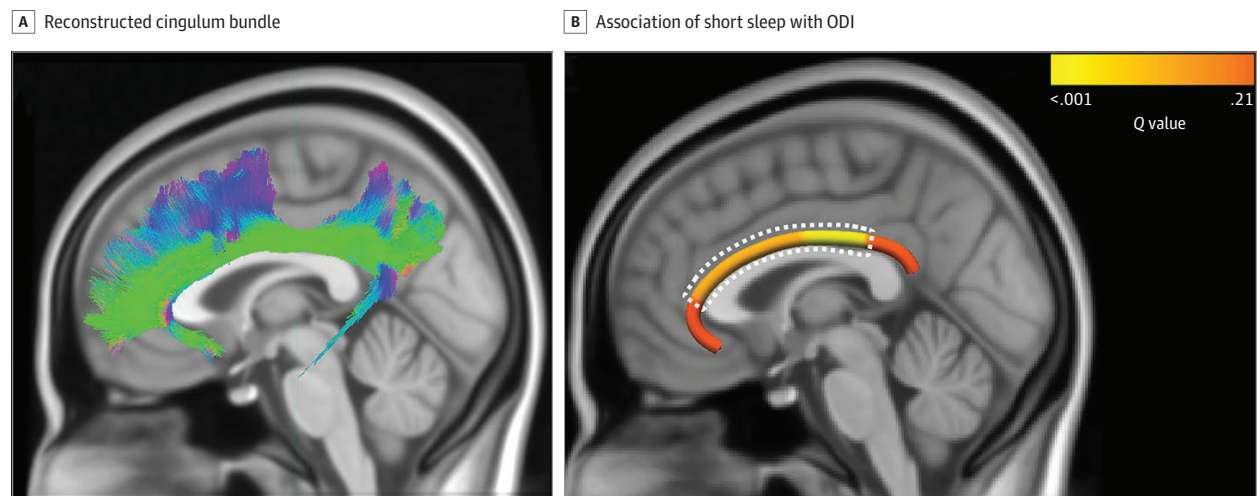
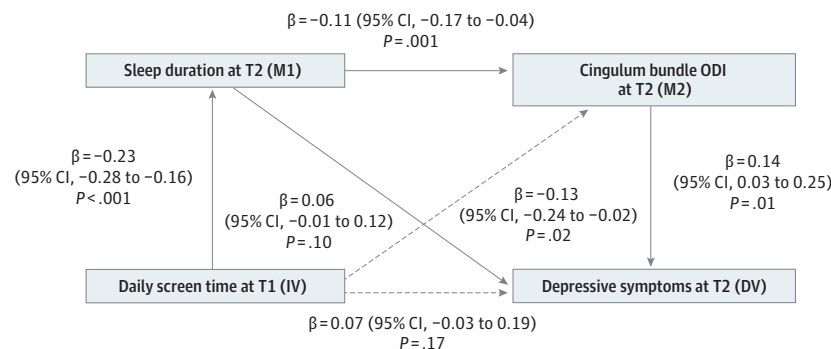


Figure 3. Serial Mediation Model Involving Screen Time, Sleep Duration, Cingulum Bundle Orientation Dispersion Index (ODI), and Depressive Symptoms



The serial mediation models involving more daily screen time, shorter sleep, greater cingulum bundle ODI, and more depressive symptoms are shown. Arrows indicate the direction of the effects and dashed lines indicate nonsignificant associations. In this model, shorter sleep mediated 37.5% (95% CI, 12.5%-50.0%) of the effects of more daily screen time on greater cingulum bundle ODI, greater cingulum bundle ODI mediated 12.5% (95% CI,

2.0%-22.7%) of the effects of shorter sleep on more depressive symptoms, and shorter sleep combined with greater cingulum bundle ODI mediated 36.4% (95% CI, 18.2%-63.6%) of the effects of more daily screen time on more depressive symptoms. DV indicates dependent variable; M1, mediator 1; M2, mediator 2; IV, independent variable.

activity. Active and passive entertainment time effects involved short sleep and worse cingulum bundle coherence, but communication time effects did not involve sleep duration. This suggests that the relationships involving social media in early adolescence may be more nuanced and components beyond time (eg, content) should be explored. Still, time may become relevant shortly, given that there has been an increase in social media use among children aged 8 to 12 years in recent years.¹ Altogether, the findings show that screen time activities can inform strategies for promoting healthy habits and improving mental health in this age group.

The cingulum bundle is the main dorsolimbic pathway and plays a role in the default mode network and emotional regulation.^{38,69} The current findings suggest that screen time, through short sleep, affects neural refinement (ODI) rather than

the formation of new connections (neurite density index), contributing to increased depressive symptoms. This aligns with research on the role of sleep in synaptic pruning and refinement.^{26,70} Moreover, the significant effect observed in the middle portion of the tract suggests that the combined impact of screen time and sleep may disrupt the integration of information across multiple brain regions and functions, potentially affecting functions beyond emotional regulation. This also aligns with the effects of screen time and sleep across multiple functions.^{6,18} It was also observed that short sleep is critical for the effects of daily screen time on white matter. Given that the threshold for significant effects of screen time on short sleep was the lowest (1 hour), the screen-sleep-brain relationship findings suggest that sleep disruption occurs before screen time impacts the brain. Moreover, the higher thresholds for

white matter organization (3 hours) align with the lower mediated proportion of 12.5% observed in the sleep-brain-depression relationship. This suggests that depressive symptoms may result from screen-related changes in other pathways (eg, psychological, cognitive, brain gray matter), which may precede detectable white matter changes and warrant further investigation. Altogether, these findings highlight the importance of closely monitoring the impact of daily screen time on sleep, because disruptions in sleep occur before screen time affects white matter and mental health, suggesting that these effects may be modifiable.

Limitations

Although this study raises important questions about the relationship between screen time and mental health, it also has limitations. The information about screen time and sleep relied on self-report, which may have introduced bias. Additionally, measurements were not applied consistently across time points. Future studies should consider using direct measurements of screen time and sleep, such as mobile sensing data and sleep actigraphy, applied consistently across all time points to better establish these relationships during this age. Although the current work highlights the impact of screen time in early adolescents, it does not consider the content

consumed. Specific types of media, such as negative or violent content, have unique effects on adolescent mental health.⁷¹ Future studies should build on these findings and examine the role of content on the sleep-brain relationship during early adolescence. Moreover, screens may be used under parental supervision in this late childhood and early adolescence, so the effect of parental monitoring style should be accounted for. This research should be expanded to other age groups to examine how changes in screen engagement throughout adolescence impact brain and mental health.

Conclusions

Findings of this study support the crucial role of sleep duration and white matter, particularly the cingulum bundle, in mediating the association between daily screen time and depressive symptoms. These findings underscore the importance of fostering healthy screen time habits and prioritizing adequate sleep to support emotional and brain development. Engaging parents, educators, and adolescents in these efforts is essential, as is further research into how screen time affects white matter, especially through mechanisms such as the influence of melatonin on the developing brain.

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Drafting of the manuscript: Lima Santos, Soehner, Ladouceur.

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